EXPERIMENT NO 14

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14.1.1 Complex number Addition using class

Algorithm

Step 1: Start

Step 2: Define a class Complex

Step 3: Define a method initComplex()

- Input two integers (real and imaginary parts)
- Store them in self.real and self.imag

Step 4: Define a method sum(c1, c2)

- Add real parts → self.real = c1.real + c2.real
- Add imaginary parts → self.imag = c1.imag + c2.imag

Step 5: Define a method display()

- If imaginary part ≥ 0 → print real + imag i
- Else → print real - |imag| i

Step 6: Create three objects: c1, c2, c3

Step 7: Call initComplex() for c1 and c2 to take input

Step 8: Call sum(c1, c2) using object c3

Step 9: Call display() using object c3

Step 10: Stop

Flowchart

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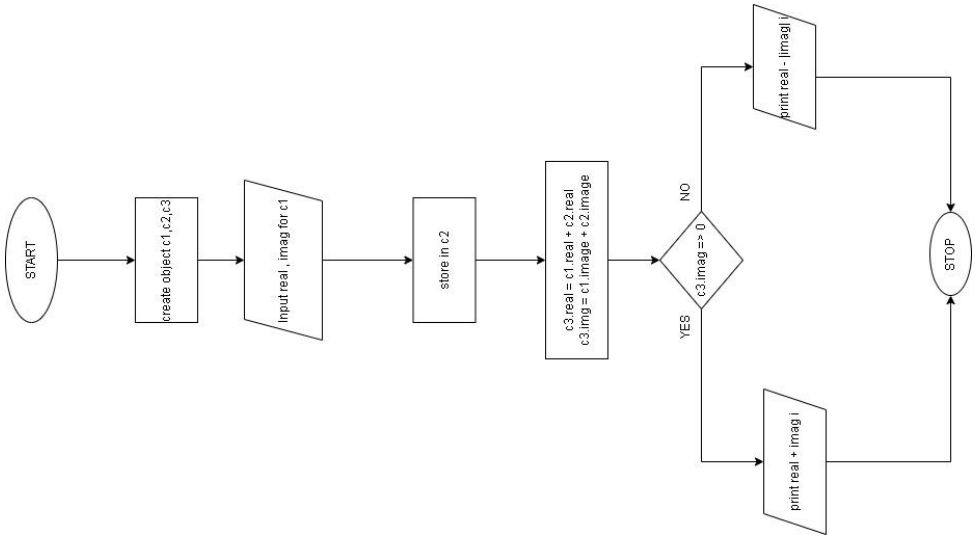

Python code

```
class Complex:
        def initComplex(self):
                self.real, self.imag = map(int, input().split())

        def sum(self, c1, c2):
                self.real = c1.real + c2.real
                self.imag = c1.imag + c2.imag

        def display(self):
                if self.imag >= 0:
                        print(f"{self.real} + {self.imag}i")
                else:
                        print(f"{self.real} - {abs(self.imag)}i")

# Create three instances
c1 = Complex()
c2 = Complex()
```

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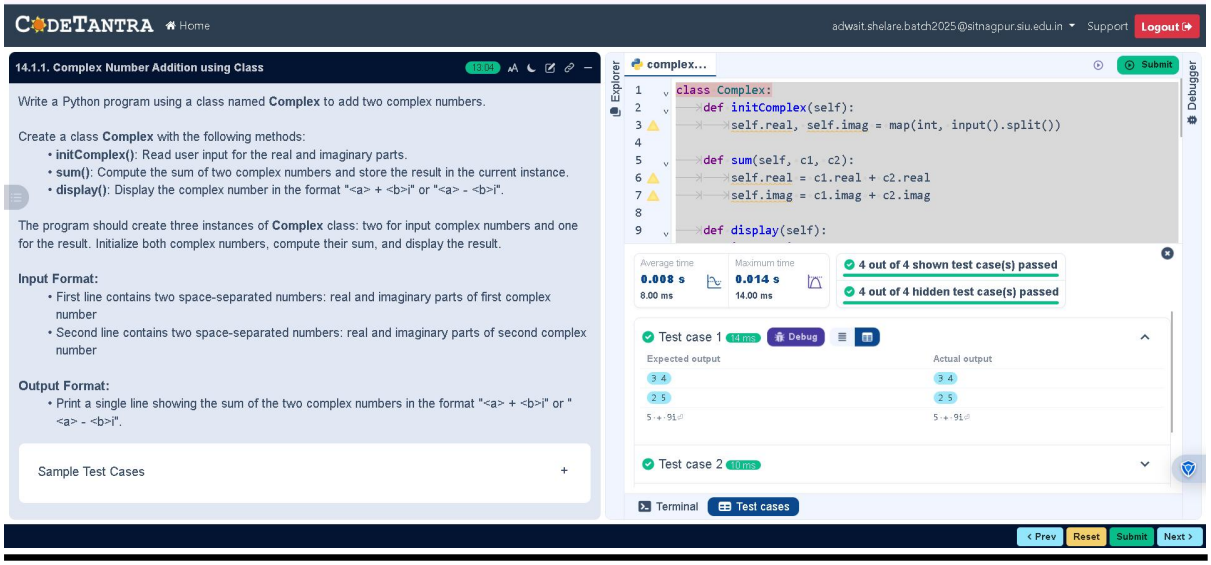
```
c3 = Complex()
# Initialize two complex numbers
c1.initComplex()
c2.initComplex()
# Compute and display sum
c3.sum(c1, c2)
c3.display()
```

EXCECUTION